

be robust enough to generalise to hypergraphs that miss a significant proportion of the edges. Moreover, the proof of the graph case ($r = 2$) from [18] relies on a good understanding of matchings in graphs in form of the Tutte–Berge formula, no analogue of which is available for hypergraphs. Finally, one could try to prove the existence of certain small coloured configurations (“gadgets”) that can be repeatedly removed until the remaining hypergraph has a very specific structure. For instance, in the case of two colours, in [17] it is implicitly proved that either there exist two edges of different colour that share $r - 1$ vertices, or the colouring is almost monochromatic. For general q , one possible gadget would be a set of $r + q - 1$ vertices that contains an edge of each colour. However, there are constructions where all colour classes are large yet there is no such gadget, so this approach seems infeasible as well. We instead follow a new approach and make use of tools from extremal set theory developed in the study of the Erdős matching conjecture (see the overview in Section 2 for more details).

Theorem 1.2 also has implications to the discrepancy of perfect matchings in random hypergraphs. The general question in discrepancy theory is whether, given a ground set Ω , a family $\mathcal{P} \subseteq 2^\Omega$ and a positive integer $q \geq 2$, there exists a q -colouring of Ω such that each set in $\mathcal{P}$ contains roughly the same number of elements of each colour. In the setting of (hyper)graphs, the main interest is finding conditions on a hypergraph G under which any q -edge-colouring of G contains a particular substructure with high discrepancy, meaning that significantly more than a $1/q$ -proportion of the substructure’s edges are in the same colour. The question goes back to works of Erdős and Spencer [12], and of Erdős, Füredi, Loeb and Sós [11]. Recently, this has been extensively studied for minimum degree conditions forcing perfect matchings with high discrepancy in hypergraphs (see [3, 17, 20, 28]), with the graph case having been considered earlier (see [2, 15, 19]). Hàn, Lang, Marciano, Pavez-Signé, Sanhueza-Matamala, Treglown and Zárate-Guerén [20, Section 5.3] showed that there exists a constant $C > 0$ such that if $p \geq C\sqrt{n^{2-r}}$, then w.h.p. in every q -colouring of the edges of $\mathbb{G}^{(r)}(n, p)$ there is a perfect matching with high discrepancy. However, due to the celebrated result of Johansson, Kahn, and Vu [22], we know that a perfect matching already typically exists for $p \gg n^{-r+1} \log n$ (cf. Theorem 3.5). A natural question, which was asked in [20], is to determine the correct threshold (depending on r, q) for the property of having a perfect matching with high discrepancy in every q -colouring. Motivated by this, we prove the following result, which starts working at the threshold for the existence of a perfect matching, and even yields an asymptotically optimal bound on the discrepancy. However, note that we assume the colouring is known a priori, that is, the statement we prove to hold w.h.p. is for a fixed colouring.

Theorem 1.4. *For all $r, q \in \mathbb{N}$ with $r, q \geq 2$, and all $\mu > 0$, there exists $C > 0$ such that, provided $p \geq Cn^{-r+1} \log n$, the following holds. For any q -colouring of the k -subsets of $[n]$, the random hypergraph $G \sim \mathbb{G}^{(r)}(n, p)$ contains w.h.p. a perfect matching with at least $(1 - \mu)\frac{n}{r+q-1}$ edges of the same colour.*

Organisation. The rest of the paper is organised as follows. The proof of the defect theorem (Theorem 1.3) is discussed in Section 2 and the proof of the transference theorem (Theorem 1.2) in Section 3. The proof of Theorem 1.4 is also discussed there. Finally, we give concluding remarks in Section 4. The proof of the multicolour weak sparse hypergraph regularity lemma is added for completeness in Appendix A.

Notation. We use standard graph theory notation. We let $K_n^{(r)}$ denote the complete r -graph on n vertices.

We let $[n]$ denote the set $\{1, \dots, n\}$ and, given a set X and an integer $i \geq 0$, we write $\binom{X}{i}$ for the collection of all subsets of X of size i .

For $a, b, c \in (0, 1]$, we write $a \ll b \ll c$ in our statements to mean that there are increasing functions $f, g : (0, 1] \rightarrow (0, 1]$ such that whenever $a \leq f(b)$ and $b \leq g(c)$, then the subsequent result holds.